Neural Data Privacy and the Emerging Surveillance Risk of Brain-Computer Interfaces

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Abstract

Brain-computer interfaces (BCIs) represent one of the most consequential emerging technologies of the 21st century. While current regulatory frameworks focus primarily on their clinical and rehabilitative applications, a critical and underexplored risk is taking shape: the potential misuse of neural interfaces as instruments of mass cognitive surveillance. This report examines the technical mechanisms by which a BCI system can acquire, encode, and transmit neural data without explicit user consent; reviews the current inadequacy of global data privacy frameworks as applied to brain-derived signals; and proposes a tiered ethical framework for neural data governance. A particular emphasis is placed on the disproportionate vulnerability of populations in Sub-Saharan Africa and Uganda specifically, where general-purpose data protection laws provide no neural-specific protections, institutional enforcement capacity is limited, and the convergence of mobile technology growth and consumer neurotechnology diffusion creates a structurally undefended exposure surface. This report argues that neural data requires a distinct legal and ethical classification; one that protects cognitive liberty as a foundational human right before commercial BCI deployment reaches critical mass in both the Global North and, increasingly, in emerging markets.

1. Introduction

For most of human history, the contents of the mind were inaccessible to external observation. The brain's continuous electrochemical activity, the substrate of thought, emotion, intention, and identity, could not be measured, decoded, or transmitted beyond the skull that contained it. That fundamental privacy of cognition is no longer guaranteed. The rapid maturation of brain-computer interface (BCI) technology which is systems that create direct communication pathways between neural tissue and external devices, has made it technically feasible to decode motor intentions, infer emotional states, and reconstruct rudimentary semantic content from neural signals in real time [1, 2].

The dominant narrative around BCIs is one of therapeutic promise: restoring motor function in paralysed patients, enabling communication for individuals with locked-in syndrome, and treating refractory neurological disorders. Landmark clinical achievements including the BrainGate consortium's demonstration of cortical control of robotic limbs [3] and Neuralink's first-in-human PRIME trial in January 2024, which enabled a paralysed patient to control a computer cursor and achieve communication speeds exceeding prior BCI records [4, 5], represent genuine engineering breakthroughs that hold transformative medical value.

However, as BCIs begin the transition from heavily regulated clinical devices to consumer products marketed for wellness, productivity, and cognitive enhancement, a less examined risk surface is emerging. The same hardware and signal processing pipelines that decode a patient's intention to move a cursor are, in principle, capable of capturing a far broader spectrum of neural information including data the user has no intention of communicating. A 2024 survey by the Neurorights Foundation of 30 direct-to-consumer neurotechnology user agreements found that 29 of 30, when accepted, granted the company rights to sell user brain data to third parties [6]. This is not a distant risk. It is a present commercial reality operating in a regulatory vacuum.

This report examines what this means technically, legally, and ethically, with specific attention to the disproportionate vulnerability of populations in Sub-Saharan Africa, where no jurisdiction-specific neural data protection exists and institutional enforcement capacity remains severely constrained. A tiered governance framework is proposed, and the argument is made that the window for preemptive action is open, but narrowing.

2. Background: How BCIs Acquire and Process Neural Data

2.1 Signal Acquisition Architectures

BCIs acquire neural data through one of three primary electrode architectures, distinguished by their degree of neural proximity and signal resolution [7]. Understanding these distinctions is essential to understanding the spectrum of privacy risk:

- **Non-invasive systems** (e.g., electroencephalography, EEG) place electrodes on the scalp surface, capturing aggregate electrical potentials from cortical populations. Spatial resolution is low (~1 cm) and signals are heavily attenuated by scalp, skull, and dura. Consumer EEG headsets from Emotiv, Muse (InteraXon), and NeuroSky fall into this category; they are the architecture most immediately relevant to mass consumer deployment.
- **Electrocorticography (ECoG)** places electrode grids directly on the cortical surface beneath the skull, providing substantially higher signal fidelity and spatial resolution (~1 mm) without penetrating neural tissue. Currently used in epilepsy monitoring and select BCI research, ECoG represents an intermediate risk tier.
- **Intracortical microelectrode arrays (IMAs)**, such as the Utah Array and Neuralink's N1 Implant (1,024 electrodes across 64 flexible threads), insert microscale electrodes directly into neural tissue, enabling single-neuron resolution recording [4]. This modality offers the highest signal bandwidth and the greatest privacy risk surface.

Table 1 summarises the key parameters of each modality and their corresponding privacy risk profiles:

Modality	Spatial Res.	Temporal Res.	Channels	Invasiveness	Privacy Risk Profile
Non-invasive EEG	~1 cm	~1 ms	8-256	None	Emotional state, attention, cognitive load
ECoG	~1 mm	~1 ms	16-256	Subdural	Motor intent, speech, fine affective states
Intracortical (IMA)	~50 um	<1 ms	128-3072	Penetrating	Single-neuron; motor, speech, semantic content

Table 1. BCI signal acquisition architectures: technical parameters and privacy risk profiles. Sources: [7, 8, 4].

A critical observation applies to all three architectures: signal acquisition hardware records electrical potentials continuously, without inherent discrimination between 'intended' signals and ambient neural activity. The distinction between clinically relevant and incidentally captured data is made entirely downstream, in software. Regulatory frameworks that rely on software-level data minimization commitments cannot, in principle, verify whether the hardware is capturing more than declared.

2.2 Signal Processing and Neural Decoding

Raw neural signals undergo a processing pipeline before they become actionable data [9]. The key stages are: (i) amplification and analog-to-digital conversion; (ii) bandpass filtering to isolate frequency bands; delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–100 Hz); (iii) feature extraction, involving spectral power estimation, spike sorting, or event related potential analysis; and (iv) machine-learning-based decoding to map extracted features to intended actions or states.

The decoded output is only one narrow slice of what the acquired signal contains. The same recording that yields a motor intention decoder simultaneously contains information correlated with emotional arousal [10], attentional state [11], cognitive load [12], and, in higher-resolution intracortical recordings, potential correlates of semantic content during speech imagery [13]. The decision about which signals to decode and transmit is a software configuration choice, not a hardware constraint. A device sold for motor rehabilitation contains, within its raw data stream, everything needed for affective profiling, if the software is configured to extract it.

3. The Neurosurveillance Risk: Technical and Conceptual Analysis

3.1 What Neural Data Reveals

Unlike behavioural data collected by conventional surveillance systems such as location, browsing history, purchase records, neural data is generative rather than merely reflective. It does not record what a person has done; it encodes the neurobiological substrates of what they intend, feel, and think. Several research groups have demonstrated proof-of-concept neural decoding capabilities that illustrate the scope of this risk:

- **Motor intention decoding:** Hochberg et al. demonstrated that intracortical recordings could decode intended cursor movements and handwriting in paralysed patients with high accuracy [3]. Willett et al. extended this to imagined handwriting, achieving 90 characters per minute via neural decoding alone [13]. Neuralink's PRIME trial participant subsequently surpassed historical BCI communication rate records in early 2024 [4, 5].
- **Affective state inference:** Studies using consumer EEG devices have demonstrated classification of emotional valence and arousal with above-chance accuracy using machine learning, raising concerns about emotion harvesting in commercial contexts [10]. This capability is deployable today on devices already in consumer markets.
- **Speech decoding:** Chang et al. at UC San Francisco demonstrated real-time decoding of intended speech from ECoG recordings, reconstructing intelligible sentences from cortical signals alone at rates approaching natural speech [14]. Moses et al. extended these methods to a 50-word vocabulary in real time [15].
- **Covert preference detection:** The 'brain fingerprinting' paradigm, demonstrated by Farwell and Donchin, showed that P300 event-related potentials could reveal recognition of information a subject did not voluntarily disclose [16]. This underpins theoretical approaches to lie detection and ideation inference. It should be noted that P300 paradigms currently require the presentation of controlled stimuli to the user, limiting their immediate application in purely passive, ambient surveillance without an interactive interface.
- **Identity and health inference:** Neural signal patterns are individually unique biometric identifiers that cannot be changed if compromised. Additionally, neurological and psychiatric conditions produce characteristic neural signatures identifiable through EEG, creating potential for covert medical profiling without clinical consent [17].

3.2 The Commercial BCI Trajectory

The BCI market is undergoing a structural shift, moving from heavily regulated clinical contexts to consumer markets through wellness, productivity, and gaming applications [18]. Major technology

companies including Apple and Snap, have begun active research into consumer BCI applications for augmented and virtual reality interfaces [19], while Chinese employers have been documented deploying EEG-based emotional monitoring systems in workplaces without robust consent frameworks [20].

The Neurorights Foundation's 2024 survey of 30 consumer neurotechnology terms-of-service agreements found that 29 of 30, when accepted via standard clickwrap, granted the company rights to sell user neural data to third parties [6]. These are not obscure startups; they include established players in consumer wellness and education neurotechnology. The gap between the data being acquired and the protections users believe they have is, at present, enormous.

3.3 A Threat Model: From Therapeutic Device to Surveillance Infrastructure

To ground the surveillance risk in concrete terms, consider the following generalised threat model for a hypothetical consumer BCI company referred to here as MindSync, operating in a low-regulation environment:

- **Hardware deployment:** MindSync offers a minimally invasive neural device with a consumer value proposition of cognitive enhancement or wellness monitoring. The device is approved as a general wellness product, avoiding the burden of medical device classification.
- **Data acquisition:** The device acquires continuous neural data streamed via Bluetooth to a smartphone application. The terms of service, accepted via clickwrap permit data sharing with affiliated third parties for 'service improvement,' consistent with terms found in the majority of current consumer neurotechnology agreements [6].
- **Neural profiling:** Machine learning models trained on aggregated neural data from millions of users construct individualised cognitive and affective profiles; attention patterns, emotional reactivity, stress responses, and preference markers without any single inferred attribute being disclosed to the user.
- **Secondary monetisation or state access:** The neural profiles are made available to advertising networks, employer screening services, insurance actuaries, or in jurisdictions without data sovereignty protections, government intelligence agencies seeking insight into political attitudes or dissent.

This is not speculative science fiction. Each component has a direct precedent: the Cambridge Analytica scandal demonstrated psychological profiling from passive digital behavioural data [23]; the Snowden disclosures established that intelligence agencies actively sought access to commercial technology platforms [24]; and Chinese workplace deployments have shown EEG-based emotional monitoring of workers already operationally deployed [20]. While Cambridge Analytica relied on behavioural rather than neural data, it established the operational precedent for mass psychological profiling.

4. Current Regulatory and Legal Landscape

4.1 Inadequacy of Existing Data Protection Frameworks

Contemporary data protection legislation was designed for digital behavioural data. Neural data does not fit cleanly into these frameworks [17]. As Yang and Jiang (2025) note, none of the major data protection laws enacted before 2024 contained explicit coverage of neural data [25]:

- **Inferential depth:** Neural data enables inferences about internal states; intentions, emotions, beliefs, that go far beyond what GDPR or CCPA contemplated. Article 9 of GDPR identifies biometric data as a special category, but was designed for identification data (facial recognition, fingerprints), not affective or cognitive inferences from ongoing neural signals.
- **Continuous generation:** Unlike a passport scan or medical test result, neural data is generated continuously and involuntarily. The concept of a discrete, informed consent event is structurally ill-suited to continuous physiological monitoring.
- **Temporal irreversibility:** Neural response patterns cannot be revoked once disclosed. A neural profile, once constructed, constitutes permanent knowledge about an individual's cognitive architecture. There is no mechanism to 'unlearn' an adversary's model of your neural responses.
- **Jurisdictional complexity:** A user in Kampala wearing a consumer BCI device manufactured in Shenzhen and running cloud processing on servers in Virginia may be subject to the data protection laws of none, one, or all three jurisdictions. The cross-border processing pipeline systematically undermines national frameworks.

4.2 Emerging Jurisdiction-Level Responses

- **Chile (2021):** Became the world's first country to constitutionally enshrine neurorights (Law No. 21.383), protecting mental integrity and requiring the law to 'especially safeguard brain activity and the information derived from it' [26]. In 2023, Chile's Supreme Court applied this in a landmark ruling against Emotiv Inc. the first court ruling anywhere on neuroprivacy [27].
- **Colorado (2024):** Enacted HB24-1058, amending the Colorado Privacy Act to explicitly include neural data as a sensitive personal data category requiring opt-in consent [28].
- **Minnesota (2024):** Signed HF 1967 into law in May 2024, extending consumer data privacy law to neural data with both civil and criminal penalties, the first jurisdiction globally to apply criminal liability to nonconsensual neural data acquisition [29].

- **California (2024):** SB1223 amended the California Consumer Privacy Act to explicitly include neural data as sensitive personal information, extending full CCPA consumer rights to brain derived data [19].
- **UNESCO (2023):** Adopted a Recommendation on the Ethics of Neurotechnology at the 42nd General Conference, establishing normative guidance including mental privacy, cognitive liberty, and mental integrity as foundational values [30]. However, UNESCO Recommendations are non-binding.

These precedents are significant since they demonstrate political will and establish frameworks others can adapt. But their geographic coverage is profoundly uneven. The vast majority of the world has no neural-specific protection whatsoever.

4.3 The Sub-Saharan African Deficit: A Disproportionate and Unaddressed Vulnerability

Why Sub-Saharan Africa Requires Specific Attention

The framing of neural data privacy as a concern primarily for wealthy, technology-intensive societies is empirically incorrect and analytically dangerous. The combination of rapid mobile technology penetration, weak neural-specific regulatory frameworks, limited enforcement capacity, growing consumer neurotechnology markets, and structural vulnerabilities to digital colonialism makes Sub-Saharan Africa one of the highest-risk regions in the world for neural data exploitation and the least discussed in the academic literature.

As of 2024, 36 of the 55 African Union member states have enacted comprehensive data protection laws [31]. However, a critical distinction must be drawn: the existence of a general data protection law and the existence of neural-data-specific protection are not equivalent. None of the data protection laws currently in force across Sub-Saharan Africa contains provisions specifically addressing neural data, brain-derived signals, or the cognitive inferences enabled by BCI systems [25, 31]. These frameworks address what data is collected; they have no framework for addressing what neural data reveals.

Uganda's Specific Regulatory Exposure

Uganda's Data Protection and Privacy Act (2019) provides the country's primary framework for personal data governance [32]. The Act establishes categories of sensitive data including health data, biometric data, and genetic data and imposes heightened consent requirements. However, the Act's definition of biometric data is framed around identification functions (fingerprints, facial geometry, iris scans) rather than continuous physiological monitoring. Neural signals acquired for motor decoding or affective inference do not clearly fall within this definition.

Furthermore, the Act contains no provisions governing the downstream inferential use of data. A consumer BCI company could, in principle, argue that EEG recordings are neither biometric nor

health data within the meaning of the Act, while using those recordings to construct detailed psychological profiles. Additionally, Uganda's Personal Data Protection Office (PDPO) under NITAU has, since the Act's passage, operated with limited technical capacity in the domain of emerging neurotechnology [33]. Uganda's situation is not unique in the East African Community: the EAC regional framework contains no neural-data-specific provisions, and the AU's 2014 Malabo Convention has been ratified by only 15 countries and addresses neural data nowhere in its provisions [31].

The Digital Colonialism Dimension

Consumer BCI devices are designed, manufactured, and owned by companies headquartered primarily in the United States, China, and Europe. The data they collect wherever their users are located, flows to servers in those jurisdictions, governed by laws those populations had no role in designing. Users in Uganda, Kenya, or Nigeria wearing consumer EEG headsets are generating neural data that flows to servers in Virginia or Shenzhen, subject to legal frameworks they have no meaningful recourse under.

This mirrors the broader dynamics of digital surveillance capitalism documented by Zuboff (2019), but with a crucial intensification: while behavioural data extraction has generated significant concern in the Global North, the extraction of neural data from populations in the Global South has attracted almost no regulatory attention and essentially no scholarship. The neural data of African users, in the current regulatory environment, is effectively unprotected by any jurisdiction-specific legal instrument.

The Mobile Technology Exposure Vector

Mobile internet penetration reached 27% across Sub-Saharan Africa by the end of 2023 and is projected to grow significantly toward 2030 [34]. Consumer neurotechnology companies — having exhausted much growth potential in saturated Global North markets — are increasingly oriented toward emerging market expansion. The combination of mobile-first adoption, low regulatory oversight, growing consumer spending, and high-growth technology markets creates a commercially attractive environment for consumer BCI deployment in East Africa specifically.

Critically, this expansion will likely occur before any jurisdiction-specific neural data protection is enacted. The relevant comparison is to mobile financial services: Uganda was an early adopter of mobile money technology, and its regulatory frameworks were largely developed retroactively, after substantial commercial deployment. The same pattern applied to neural data would have profoundly more serious consequences, given the irreversibility of neural profiles once constructed.

The Stakes for Uganda and the Region

Uganda has no neural-data-specific law. Its general data protection framework does not clearly cover neural inferences. Its data protection regulator has no demonstrated technical capacity in neurotechnology. Consumer EEG devices are already available for purchase in Kampala. Regional harmonisation through the EAC or AU has not addressed this gap. In the absence of deliberate intervention, Uganda and most of Sub-Saharan Africa will be fully exposed to consumer neural data extraction before any protective framework exists.

5. A Proposed Ethical Framework for Neural Data Governance

Drawing on the neurorights framework formulated by Ienca and Andorno [35], the legislative precedents reviewed above, and the specific vulnerabilities identified in Section 4.3, this report proposes a four-tier governance model for neural data. The framework is explicitly designed to be applicable to both high-income regulatory environments and to resource-constrained jurisdictions in the Global South.

Tier 1: Cognitive Liberty as a Foundational Right

The right to mental self-determination; to think, feel, and form intentions without external surveillance must be recognised as a foundational human right, not merely a privacy interest. This requires legal classification of neural data as a distinct, highest-sensitivity category of personal data, explicitly separate from both medical records and behavioural data. No neural data acquisition should be permitted outside explicit, granular, revocable informed consent. Consent must address not only collection but the full range of potential inferences, and must be renewed when new decoding capabilities are added to a deployed system.

Tier 2: Purpose Limitation and Minimum Necessary Acquisition

Neural data acquisition must be governed by strict purpose limitation: data collected for motor intention decoding must not be used for affective state inference, cognitive profiling, or any secondary purpose. Regulatory bodies must mandate hardware-level data minimization, not merely software policy compliance. This requires technical standards bodies including IEEE and IEC to develop hardware attestation protocols that independently verify devices acquire only the signal bandwidth consistent with their declared function.

Tier 3: Algorithmic Transparency and Independent Auditability

Any commercial BCI system must be subject to independent algorithmic audit to verify that decoding is limited to declared functions. For resource-constrained jurisdictions, the AU and EAC should establish a shared regional neurotechnology audit body, preventing individual countries from bearing the full cost of building specialised technical capacity. Open-source reference decoder implementations should be developed and maintained as a public good.

Tier 4: International Neural Data Governance with Explicit Attention to the Global South

An international treaty framework coordinated through UNESCO, building on its 2023 Recommendation on Neurotechnology Ethics [30] is necessary to establish minimum global standards. This framework must be explicitly designed with Global South participation, not merely Global South ratification of standards developed elsewhere. African Union member states, including Uganda, should be active co-drafters of any international neural data governance instrument, not passive recipients of Northern regulatory frameworks.

6. Discussion

The central argument of this report is that the window for preemptive governance of neural data is open, but closing. Historical precedents from the digital era are instructive: the failure to regulate social media data aggregation in the 2010s created surveillance architectures extraordinarily difficult to dismantle after they were embedded in commercial infrastructure [23]. The neural case is structurally analogous, but the stakes are higher because the data being extracted is not behavioural but cognitive; not what people do, but what they think and feel.

On the innovation objection. The strongest objection to preemptive neural data regulation is that it will stifle beneficial development, particularly in therapeutic applications. This objection has merit in the clinical domain: therapeutic BCIs serving patients under IRB oversight are already subject to medical device regulatory infrastructure that provides appropriate safeguards. The objection does not have merit in the consumer domain, where the primary driver is commercial rather than therapeutic.

A technically sophisticated version of the objection argues that current consumer BCIs do not acquire the high-resolution data necessary for the threat model described in Section 3. This is correct as a statement of current capability. It is not valid as a basis for regulatory inaction: the cost trajectory of neural interface hardware is downward and the capability trajectory is upward. Regulation that waits for demonstrated harm will arrive too late. Furthermore, the Neurorights Foundation's finding that 29 of 30 current consumer neurotechnology TOS agreements already permit third-party data sale [6] demonstrates that the commercial infrastructure for neural data exploitation is being constructed now, with current-capability devices.

On grounds for cautious optimism. The field of neuroethics is more mature relative to the current state of BCIs than internet ethics was relative to the early web. The Chilean constitutional precedent and the Minnesota criminal liability model demonstrate that novel governance instruments are politically achievable. The question is whether these forces can be mobilised before commercial deployment in underprotected markets particularly in Africa establishes facts on the ground that are difficult to reverse.

Limitations of this Report

While this report identifies critical vulnerabilities in the current trajectory of consumer neurotechnology, several limitations to its scope and proposals must be acknowledged. First, the regulatory analysis is not an exhaustive global mapping; it focuses specifically on pioneering precedents in the Americas (Chile, U.S. states) and the structural deficits in Sub-Saharan Africa, particularly Uganda. Consequently, it omits detailed examination of other major regulatory regimes,

such as the European Union's AI Act or emerging data governance frameworks in the Asia-Pacific region.

Second, the threat model presented in Section 3 is illustrative and predictive. While it is grounded in the current capabilities of BCI technology and existing consumer terms-of-service agreements, it represents a projection of near-future risks rather than an empirically documented case study of mass-market neural surveillance already occurring at scale.

Finally, the four-tier ethical framework proposed in Section 5 is a normative blueprint rather than a technical implementation manual. Specifically, Tier 2 (hardware-level data minimization) faces significant engineering constraints. Because current non-invasive and invasive electrodes physically record continuous, broad-spectrum analog electrical potentials, enforcing purpose-limitation strictly at the hardware level remains an unsolved engineering challenge. Implementing this tier will require substantial future research and development into edge-AI preprocessing, physical bandpass filtering, and secure enclave architectures that do not currently exist in consumer devices.

7. Conclusion

Brain-computer interfaces are not merely a new category of medical device or consumer gadget. They represent the first technology capable of systematic access to the substrates of human thought. The engineering principles that make BCIs therapeutically valuable; high-fidelity neural signal acquisition, real-time decoding, and wireless transmission are the same principles that make them capable of unprecedented cognitive surveillance.

The technical capability for neural decoding already exists and is expanding. The regulatory frameworks adequate to govern it do not exist in most of the world. This gap is most acute in Sub-Saharan Africa, where general data protection laws provide no neural-specific coverage, enforcement capacity is limited, and the mobile-first technology adoption pathway creates structural vulnerability to early consumer BCI deployment ahead of any protective framework.

Filling this gap requires: recognition of cognitive liberty as a foundational right in both national and international law; mandatory hardware-level data minimisation with independent technical attestation; algorithmic auditability for commercial BCI systems, including a regional mechanism for the AU and EAC; and international treaty coordination on neural data governance standards designed with genuine Global South participation.

The most important insight from the history of digital surveillance is that commercial infrastructure for data extraction, once constructed at scale, is extraordinarily difficult to dismantle. The BCI industry has the rare opportunity to build an ethical foundation before the architecture is set. That opportunity is particularly urgent for populations in Sub-Saharan Africa, who have the most to lose from a regulatory failure and the least institutional voice in current governance discussions.

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